Day-37

Network Time Protocol

The importance of time

- All devices have an internal clock (routers, switches, your PC etc)
- On Cisco IOS, we can view the time with show clock command.
- The default time zone is UTC. (Coordinated Universal Time)
- If we use the 'show clock detail' command, we can see the time source

```
R1 # show clock detail
*00:19:49.411 UTC Sat Dec 26 2020
Time source is hardware calendar.
```

* = time isn't considered authoritative. (device has no confidence that its accurate)
The hardware calendar is the default time source.

- The internal hardware clock of a device will drift over time, so its not the ideal time source.
- From a CCNA perspective, the most important reason to have accurate time on devie is to have accurate logs for troubleshooting.

show logging

Suppose we logged to R2 and see time.

R2 # *01:17:06.706 Dec 27 2020.

Now we did from R3

R3 # ~~15:30~~ 16:30:37.020 UTC Fri May 23 2005

Now, think we had opened Routers to check logs and timestamp on them is totally different. So, its very important to have same time

Manual Time configuration

- We can manually configure the time on the device with the "clock set" command.

```
R2 # clock set ?
  hh:mm:ss  current time
R2 # clock set  14:30:00 27 Dec 2020.
```

All of them are done from privileged exec mode. Now after we've set time. and see time with "show clock detail" command the time source is user configuration.

- Although the hardware calendar (built-in clock) is the default time-source the hardware, clock and software clock are separate and can be configured separately.

Hardware clock configuration

- We can manually configure the hardware clock with the calendar set command. The options after calendar set are same as clock set.
- To see hardware clock use the below command.

```
R2 # show calendar
14:35:07 UTC Sun Dec 27 2020
```

- Typically we'll want to synchronize the 'clock' and calendar.
- Use the command "clock update-calendar" to sync the calendar to the clock's time
- Use the command "clock read-calendar" to sync the clock to the calendar's time.

Scanned with CamScanner

Configuring time-zone

- We can configure the time-zone with the 'clock timezone' command while configuring timezone, after keyword timezone u'us use the name to denote your timezone & its offset (diff from GMT / UTC.

Eg:

R2 (config) # clock timezone JST 9

Daylight Saving Time (Summer Time)

Not all countries do this, countries like Canada etc do this. Tgo 2 bada 3 bajney wala

- Canada set clk forward one hour on the second sunday of March at 2 AM. and then sets the clock back one hour on the first Sunday of November at 2 AM

- The command to configure this is below

R2 (config) # clock summer-time EDT recurring 2 Sunday March 02:00 1 Sunday November 2:00

So, 2 means 2nd week of March on Sunday & 1 means 1st Sunday of November.

We use recurring instead of date so that it can work ~~forever.~~ automatically every year.

Network Time Protocol

- Manually configuring the time on devices isn't scalable.
- The manually configured clocks will drift, resulting in inaccurate time.
- NTP (Network Time Protocol) allows automatic syncing of time over a network

CMD.

```
C:\Users>nslookup time.windows.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    time.microsoft.akadns.net
Address:  20.43.94.199
Aliases:  time.windows.com
```

- NTP clients request the time from NTP servers
- A device can be an NTP server and an NTP client at same time.
- NTP allows accuracy of time within ~1 milliseconds if the NTP server is in the same LAN, or within ~50 milliseconds if connecting to NTP server over a WAN/ the Internet
- Some NTP servers are 'better' than others. The 'distance' of an NTP server from the original reference clock is called stratum
- NTP uses UDP port 123 to communicate.

Scanned with CamScanner

Reference Clocks

- A reference clocks is usually a very accurate time device like an atomic clock or GPS clock.

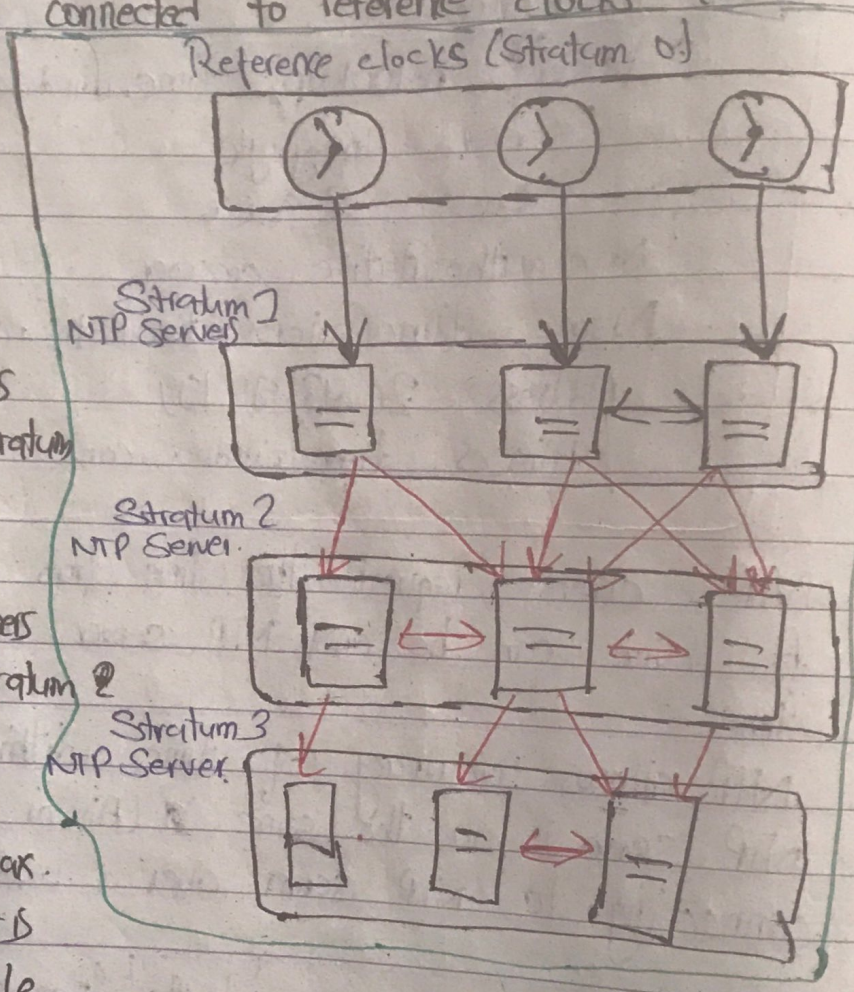
- Reference clocks are stratum 0 within the NTP hierarchy
- NTP servers directly connected to reference clocks are stratum 1.

- Reference clocks are stratum 0.

- Stratum 2 NTP servers get their time from stratum 1 NTP servers

- Stratum 3 NTP servers get their time from stratum 2 NTP servers

- Stratum 15 is the max. Anything above that is considered unreliable.



- Devices can also 'peer' with devices at the same stratum to provide more accurate time. This is called 'symmetric active' mode.
- Cisco devices can operate in 3 NTP modes:
  - Server mode.
  - Client mode
  - Symmetric active mode

- An NTP client can sync to multiple NTP servers
- NTP servers which get their time directly from reference clocks are also called as primary servers.

- NTP Servers which get their time from other NTP Servers are called secondary servers. They operate in server mode & client mode at the same time.

Network Topology

